Practice Problem Set 2

ECON 101 – Macroeconomics

Fall 2026

1. The market for coffee is initially in equilibrium. Suppose research shows that drinking coffee reduces the risk of heart disease.
   (a) Illustrate on a graph the impact on the demand curve.
   (b) Show the new equilibrium price and quantity.
   (c) Explain the changes.
2. In the market for smartphones, a new technology reduces production costs.
   (a) Draw the original supply and demand curves.
   (b) Show how the supply curve shifts.
   (c) Indicate the effect on equilibrium price and quantity.
3. Suppose demand and supply in the market for tablets are given by:

$$Q_d = 100 - 2P \qquad Q_s = 20 + 3P$$

   where Q is quantity and P is price in dollars.
   (a) Find the equilibrium price and quantity.
   (b) Determine whether there is a shortage or surplus if $P = 10$.
4. Consider the same market as in Problem 3. Suppose the government imposes a price floor of $P = 20$.
   (a) Calculate the quantity demanded and supplied at this price.
   (b) Determine the size of the surplus.
   (c) Is this price floor binding? Why or why not?

5. In the housing rental market, demand and supply are given by:

$$Q_d = 80 - 2P \qquad Q_s = 10 + 2P$$

If the government sets a rent ceiling at $P = 15$:

(a) Find the equilibrium price and quantity without the ceiling.

(b) Compute the shortage created by the ceiling.

(c) Discuss one possible unintended consequence of this policy.

6. **Expenditure Approach**
Suppose in 2024, the following data for Canada are reported (in billions of dollars):

- Consumption = 1,400
- Investment = 500
- Government Purchases = 600
- Exports = 400
- Imports = 450

(a) Calculate GDP using the expenditure approach.

(b) If depreciation is $200 billion and net factor payments from abroad are –$50 billion, calculate **Net National Product (NNP)** and **Gross National Product (GNP)**.

7. **Income Approach**
A simplified economy reports the following for 2024 (in billions of dollars):

- Wages and Salaries = 2,200
- Rental Income = 300
- Corporate Profits = 400
- Net Interest = 200
- Indirect Taxes less Subsidies = 250
- Depreciation = 150

(a) Compute GDP using the income approach.

(b) Compare it to the expenditure method if GDP was $3,350 billion. Are there any statistical discrepancies?

8. **Real vs Nominal GDP**
A two-good economy produces only bread and smartphones.

Year	Bread (Qty)	Price	Smartphones (Qty)	Price
2023	100	$2	50	$400
2024	110	$2.20	60	$380

(a) Compute **Nominal GDP** for 2023 and 2024.

(b) Using 2023 as the base year, compute **Real GDP** for 2023 and 2024.

(c) Calculate the **GDP deflator** for 2024.

9. **CPI Calculation**
Assume a typical consumer basket contains:

- 10 loaves of bread
- 5 movie tickets

Prices are given below:

Year	Bread Price	Movie Ticket Price
2023	$2	$10
2024	$2.20	$12

(a) Compute the **CPI** in 2023 (base year) and 2024.

(b) Calculate the **inflation rate** between 2023 and 2024.

(c) Compare the CPI inflation rate with the GDP deflator inflation rate from Question 3. Why might they differ?

10. **Nominal vs Real Wages**
A worker earns $20 per hour in 2023. The CPI rises from 100 in 2023 to 110 in 2024.

(a) If the worker's nominal wage in 2024 is $21, compute the **real wage** in 2024 (base year = 2023).

(b) Did the worker experience an increase in purchasing power?